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Comparison of dilution factors for German wastewater treatment plant effluents in receiving streams to the fixed dilution factor from chemical risk assessment

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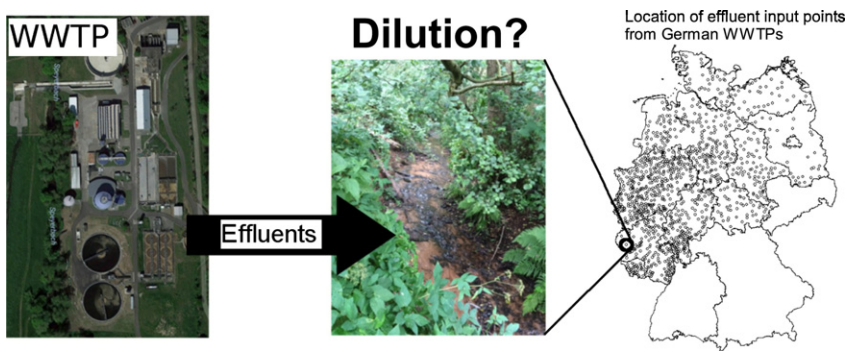
HIGHLIGHTS

- Dilution factors for German WWTP effluents were calculated for two flow conditions.
- Fixed dilution factor of 10 used in chemical risk assessment was evaluated.
- >40% of empirical dilution factors are 2-fold lower than the fixed dilution factor.
- 60% of WWTPs have at least one WWTP upstream the receiving water body.

GRAPHICAL ABSTRACT



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ABSTRACT

Incomplete removal during wastewater treatment leads to frequent detection of compounds such as pharmaceuticals and personal care products in municipal effluents. A fixed standard dilution factor of 10 for effluents entering receiving water bodies is used during the exposure assessment of several chemical risk assessments. However, the dilution potential of German receiving waters under low flow conditions is largely unknown and information is sparse for other European countries. We calculated dilution factors for two datasets differing in spatial extent and wastewater treatment plant (WWTP) size: a national dataset comprising 1225 large WWTPs in Central and Northern Germany and a federal dataset for 678 WWTPs of a single state in Southwest Germany. We found that the fixed factor approach overestimates the dilution potential of 60% and 40% of receiving waters in the national and the federal dataset, with median dilution factors of 5 and 14.5, respectively. Under mean flow conditions, 8% of calculated dilution factors were below 10, with a median dilution factor of 106. We also calculated regional dilution factors that accounted for effluent inputs from upstream WWTPs. For the national and the federal dataset, 70% and 60% of calculated regional dilution factors fell below 10 under mean low flow conditions, respectively. Decrease of regional dilution potential in small receiving streams was mainly driven by the next WWTP upstream with a 2.5 fold drop of median regional dilution factors. Our results show that using the standard dilution factor of 10 would result in the underestimation of environmental concentrations for authorised chemicals by a factor of 3–5 for about 10% of WWTPs, especially during low flow conditions. Consequently, measured environmental concentrations might exceed predicted environmental concentrations and ecological risks posed by effluents could be much higher, suggesting that a revision of current risk assessment practices may be required.

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1. Introduction

Effluents of municipal wastewater treatment plants (WWTP) are major sources for organic pollutants in the receiving water bodies (Daughton and Ternes, 1999; Wittmer et al., 2010). These pollutants include a wide range of industrial and consumer chemicals, such as additives, pharmaceuticals and personal care products (PCPs) as well as biocides (Schwarzenbach et al., 2006). In addition to the concentration in raw wastewater, the load of these micropollutants in the effluent is driven by the removal efficiency of WWTPs, which have been primarily designed to reduce the load of macropollutants, such as nutrients and organic pollution. Treated wastewater largely meets current water quality standards, but the removal of many pharmaceuticals is incomplete (Loos et al., 2013). Indeed, the removal efficiency for a variety of high-prescription pharmaceuticals varies strongly and depends largely on the treatment technique as well as the physico-chemical properties of a compound (Michael et al., 2013; Petrie et al., 2015; Tadkaew et al., 2011). For example, the effluent concentrations of compounds like carbamazepine and sulfamethoxazole remain almost unchanged in activated sludge treatment (Clara et al., 2005; Joss et al., 2005). Consequently, both substances were frequently detected in effluents (89% and 81%, respectively), beside many other pharmaceuticals (Loos et al., 2013; Miège et al., 2009). In contrast, many PCPs, such as antimicrobials, sunscreen-agents, surfactants or preservatives are generally well removed in WWTPs, but their widespread use results in high concentrations in raw wastewaters and notable concentrations remain in the effluent (for example: Triclosan, mean effluent concentrations 25 μg L^{-1}, 4-benzophenone, mean concentrations: 2701 μg L^{-1}) (Kasprzyk-Hordern et al., 2009). A study on effluents of five WWTPs in Spain showed that pharmaceuticals and PCPs (PPCPs) contribute up to 50% of the total load of investigated contaminants, and daily loads of 1 kg and 3 kg of PPCPs were discharged into receiving waters at two British WWTPs (Bueno et al., 2012; Kasprzyk-Hordern et al., 2009; Petrie et al., 2015). In addition to incomplete removal during wastewater treatment, some pharmaceuticals show high persistence in aquatic sediment systems with DT_{90}-values above 365 d (DT_x = degradation time of x% of a substance) and mean DT_{50}-values of 19 and 82 days in a mesocosm study for sulfamethoxazole and carbamazepine, respectively (Lam et al., 2004; Löffler et al., 2005). Even for relatively fast degrading compounds with low DT_{50} values (e.g. Ibuprofen, DT_{50} = 4.6 d), the high loads entering surface waters can lead to pseudo-persistence in aquatic environments, which means that their emission rates considerably exceed environmental removal rates (Bu et al., 2016; Fono et al., 2006; Osorio et al., 2016).

Several studies have demonstrated adverse effects of organic pollutants from WWTPs on aquatic ecosystems downstream of effluent input points. For example, a study analysing monitoring data from 247 sites at central German streams, found that ecological quality metrics were significantly lower when a wastewater treatment plant was within a 3 km upstream distance (Bunzel et al., 2013). Similarly, a multi-year study found a decrease in an ecological quality metric downstream from a WWTP and suggested a recovery of the community after the implementation of an additional treatment step (Ashauer, 2016). Two studies found elevated risk quotients (RQ > 1) and increased response to several toxicological endpoints at sampling sites downstream of Swiss WWTPs where dilution factors (DF) were below 5 (Munz et al., 2016; Neale et al., 2017). Finally, a reduced dilution of WWTP effluents during a low flow period was associated with a stronger effect on macroinvertebrate communities and ecosystem functioning compared to mean flow conditions (Englert et al., 2013). Generally, the dilution of the effluent volume determines the concentration in the receiving water body and is therefore a major determinant of the ecological risks from WWTP effluent.

Consequently, the ecological risk assessment of many chemical regulations, e.g. for medicinal products for human use or for biocidal products, employs a fixed standard DF_{fix} of 10 when evaluating chemical exposure of running waters via the WWTP route (ECHA, 2016; EMA, 2006). Overestimating the dilution potential of receiving waters in the exposure assessment could undermine the protectiveness of these risk assessments and potentially result in concentrations in the receiving waters that exceed the predicted environmental concentration (PEC) of the exposure models used. However, a comprehensive analysis of the real dilution potential of receiving streams is lacking. Most previous studies calculated DF for only few WWTPs (Gros et al., 2010; Kasprzyk-Hordern et al., 2009) or modelled effluent and discharge volumes to derive average annual DF (Keller et al., 2014). The minimum DF of six studies ranged from 1.1 to 63 for WWTPs from different European countries (e.g. Switzerland, Spain, United Kingdom, Italy), with approximately one quarter of the reported values below 10. The lowest reported DF (i.e. 1.1 and 3.1) were reported for each of the two WWTPs in the Swiss studies, corresponding to the smallest receiving streams covered in the mentioned European studies (Ort and Siegrist, 2009; Vermeirssen et al., 2006). The minimum DF for five WWTPs from Spain ranged from 5 to 63, with only one DF below 10 (Gros et al., 2007). No DF below 10 were reported for WWTPs from Italy (one WWTP, minimum DF = 19, Verlicchi et al., 2014) and the United Kingdom (four WWTPs, DF from 15.2 to 57.7, Baker and Kasprzyk-Hordern, 2013; Kasprzyk-Hordern et al., 2009).

Based on modelled stream flow rates for larger water bodies and WWTP effluent volumes, the estimated annual median DF for Germany is 32.3, with a range from approximately 10 to 130 (Keller et al., 2014). A modelling study by De Greef and De Nijs (1990) for Dutch surface waters predicted DF below 5 for up to 20% of small streams. Based on modelled stream data and aggregated effluent volumes (*EV*) median DF at low flow conditions between 5 and 25 were predicted for the USA (Kapo et al., 2015). Based on these studies, a relevant fraction of surface waters might exhibit DF below 10, particularly during dry seasons with low flow. However, the DF reported in previous studies have either been modelled or represent only limited temporal and spatial scales. Hence, given the non-quantified uncertainties for previous models and for extrapolating to different scales, studies with empirical DFs are required to evaluate potential risks for German streams.

Therefore, we calculated DF for the effluents of WWTPs entering German surface waters using mean low flow (MLF) conditions, a long term measure of stream flow and estimated the fraction of sites, where the DF were below the standard DF_{fix} of 10. By using the MLF of the receiving waters the calculated DF are based on reoccurring flow conditions, providing reliable insights in the dilution potential of German streams. Our calculations were based on two datasets: First, data compiled for 1225 WWTPs with at least 10,000 population equivalents (PE) and MLF of the related receiving waters covering Central and North Germany. The second dataset provided data on 708 WWTPs with at least 50 PE of one federal state in southwest Germany with related mean flow rate (MF) and MLF. The first dataset can be considered representative for large WWTPs on the national scale, whereas the second dataset complemented the analysis with smaller WWTPs and an additional flow metric. We calculated DF for each single WWTP and for two regional scenarios, where we accounted for effluents from upstream WWTPs.

2. Materials and methods

2.1. National dataset

The national dataset comprised data from 2010 and 2011 on 1381 WWTPs with a design capacity of at least 10,000 PE from 13 of the 16 German federal states (excluding Bavaria, Baden Württemberg and Saxony for which no data were available, Fig. 1a). Within the framework of the Urban Waste Water Treatment Directive 91/271/EEC federal states of Germany provided data on German WWTPs and were asked by the German Environment Agency (UBA) to additionally provide information on flow rates of respective receiving waters. The UBA merged WWTP data and respective flow rates and provided this dataset for

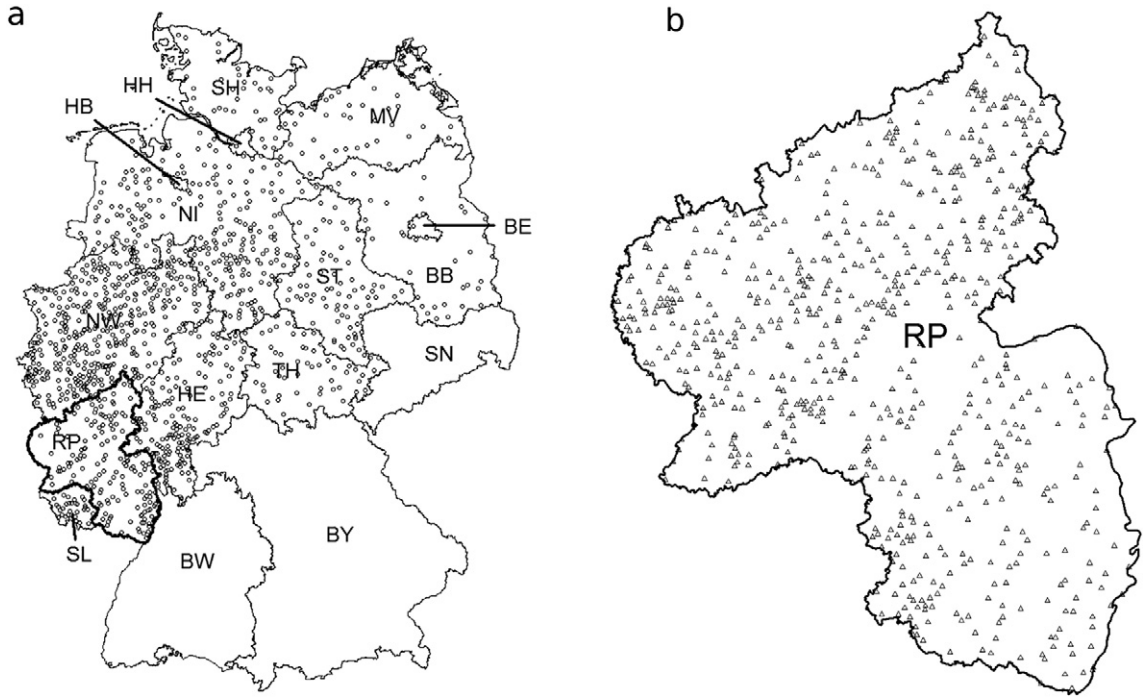

Fig. 1. Locations of effluent input points of the national dataset (a) and the federal dataset (Rhineland Palatinate, b), BB = Brandenburg, BE = Berlin, BW = Baden-Württemberg, BY = Bavaria, HB = Bremen, HH = Hamburg, HE = Hesse, MV = Mecklenburg-Vorpommern, NI = Lower Saxony, NW = North Rhine-Westphalia, RP = Rhineland Palatinate, SH = Schleswig Holstein, SN = Saxony, SL = Saarland, ST = Saxony-Anhalt, TH = Thuringia.

further analysis. For simplicity, we refer to these data as national scale, though Southern Germany is incompletely covered.

The dataset included for each WWTP *i* information on coordinates of effluent input points, nominal hydraulic loads in terms of PE_i, annual effluent volume (EV_i, m^3 $year^{-1}$), river basin district and receiving water body with stream flow rate Q_i, reported as the MLF_i (in m^3 s^{-1}), where MLF is defined as the arithmetic mean of the lowest annual flow rates over a long time period (i.e. >30 years).

Data pre-processing comprised checking for availability of information (at least EV_i, PE_i and MLF_i) per *i* that are required for our analysis. WWTPs with insufficient information were omitted from the analysis. The final dataset consisted of 1225 WWTPs (Fig. 1a), omitting 156 WWTPs lacking the required information.

2.2. Federal state dataset

The federal state dataset refers to the German federal state Rhineland-Palatinate and consists of four individual datasets provided by the State Agency for Environment. The first dataset comprised data from 2010 on 708 WWTPs with a design capacity of at least 50 PE (compare Fig. 1), and included information on PE and coordinates of effluent input points. In addition, the State Agency for Environment provided three datasets consisting of the identical stream sections with associated sub-catchments and information on a) *EV*, b) *Q* as MLF and c) *Q* as MF, where MF is defined as the arithmetic mean of the daily flow rate over a certain time period. All information included in these three datasets originated from the years 1979 to 1998.

In a first step, we merged the information from these three datasets, resulting in stream sections with related sub-catchments and *EV*, MLF and MF (where available) as attributes. Subsequently, we combined these data with the dataset on WWTPs to add the attributes (i.e. *EV*, MLF and MF) to the individual WWTPs. This was done through matching the sub-catchments with coordinates of effluent input points. After processing, approximately 14% of the WWTPs lacked *EV*. This mismatch could result from (1) incorrect coordinates of the effluent input point (resulting in the allocation to an incorrect stream section) or (2) missing *EV* information of the stream sections. Given that we could not clarify which of the two reasons were responsible, we assigned these *EV*s information of WWTPs from the national dataset based on WWTP identifier, where available. To avoid potential misallocations, we checked the plausibility of these assignments by comparing the *EV* and the MLF after standardizing to the same unit (m^3 s^{-1}). For 4.2% of the WWTPs in the federal state dataset the reported *EV* exceeded the respective MLF and we excluded these data to increase the reliability of our analysis. The final dataset comprised 678 WWTPs *i* with complete information on EV_i, PE_i, MF_i and MLF_i.

2.3. Calculation of dilution factors

The calculation of the DF for a given WWTP *i*, depends on whether the corresponding stream flow rate Q_i (either MLF or MF) excludes or includes the effluent volume for *i* (EV_i). If the stream flow rate Q_i excludes EV_i, which equates to the flow rate measured upstream from the effluent input point, the DF_{ex} is calculated according to Eq. (1):

$$DF_{ex,i,j} = \frac{\sum_{k=i}^{i+j} EV_k + Q_i}{\sum_{k=i}^{i+j} EV_k} \quad (1)$$

where *j* is the number of WWTPs *k* (with the maximum *j* defined as *n*, corresponding to the total number of upstream WWTPs) that are located upstream of the WWTP *i*.

Consideration of WWTPs discharging into tributaries within the catchment upstream of *i* was only possible for the federal state dataset, as the national dataset lacked catchment information. Hence, for the federal state dataset, we included all WWTPs of the upstream catchment related to *i*, whereas for the national dataset, only WWTPs with identical receiving water body names as for *i* were used in the calculation of regional DF. Note that this is only relevant for scenarios II) and III) described below.

If the stream flow rate includes *EV*, which equates to the flow rate measured downstream from the effluent input point, the DF_{in} is

calculated according to Eq. (2):

$$DF_{in,i,j} = \frac{Q_i}{\sum_{k=i}^{i+j} EV_k} \quad (2)$$

We calculated the $DF_{ex,i,j}$ and $DF_{in,i,j}$ for three scenarios:

I) Local scenario with $j = 0$. This is referred to as the local DF omitting any catchment influence, as for example suggested by the ECHA guidance document on environmental exposure assessment (ECHA, 2016), given as $DF_{i,0}$ hereafter (see below for details).
II) Regional scenario with $j = n$. This is referred to as the regional DF. As mentioned above, some chemicals, such as many pharmaceuticals, exhibit high environmental persistence, which may lead to the accumulation of concentrations in the water phase along a water body in case of several point sources. To account for this scenario, we assumed that the *EV* input along a stream aggregates, i.e. negligible degradation occurs between *EV* input points, denoted as $DF_{i,n}$ hereafter (see below for details).
III) Stepwise regional scenario. This is computed for the WWTP *i* with the maximum stream flow rate within a catchment or along a receiving water body (denoted $WWTP_{down}$), and all other WWTPs upstream of $WWTP_{down}$ are sorted in descending order, i.e. that the WWTP $i + 1$ is the WWTP with the second highest stream flow rate and the WWTP *n* is the WWTP with the lowest flow rate. This scenario was used to calculate the stepwise regional DF to illustrate the influence of the number of WWTPs on the DF, hereafter denoted as $DF_{down,j}$ (see below).

As a worst case assumption and for purpose of consistency, we assumed that *Q* included *EV*. Hence, Eq. (2) (DF_{in}) was used to calculate DF for the national dataset. However, for approx. 25% of WWTPs of the national dataset, the reported *EV*s were higher than the related MLF, resulting in implausible DF (i.e. DF < 1). Although the *EV* is related to the mean effluent volume over a year while the MLF to the mean low flow conditions of a period of days or weeks over decades, and consequently the reported *EV* of the considered year might indeed be higher than the MLF, we assumed in these cases that *Q* excluded the *EV* and used instead Eq. (1) (DF_{ex}) for the calculation to increase the reliability of our analysis. Note that all resulting DF_{ex} for these 25% of WWTPs were necessarily lower than 2 ($1 < DF_{ex} < 2$), since for these cases EV_i exceeded Q_i. Only 10 WWTPs of the national dataset for which *EV* > MLF matched the federal state dataset. Using the federal state MLF would lead to $DF_{i,0} > 1$ for 4 of the 10 cases, suggesting that indeed for a relevant fraction of WWTPs $DF_{i,0} < 1$ are implausible. Given that the DF calculations for the national dataset included DF_{in} and DF_{ex} for all three scenarios and for reason of simplicity, we omitted "in" and "ex" in the notation of the various DF: $DF_{i,0}$, $DF_{i,n}$, and $DF_{down,j}$.

2.4. Data analysis

We used the nominal load (i.e. PE) as proxy for WWTP size in our analyses, because (1) the given design capacity generally differed from the actual connected PE for the national dataset and (2) no design capacities were given in the federal dataset. Thus, nominal load in PE was the only basis to compare WWTP-sizes across datasets.

The statistical software R was used for all calculations (R Development Core Team, 2016). Due to non-normal distribution of DF, the non-parametric Kruskal-Wallis test (KW-test) was used to test for significant differences in the DF between federal states from the national dataset. We set the null hypothesis H_0: The central tendency of DF is the same across federal states in the national dataset. If H_0 is true or the differences between the central tendencies of DF are minor, the results for Rhineland Palatinate can be considered as representative for the other German federal states. Due to small sample sizes, the federal states Berlin, Bremen and Hamburg (n = 3, 4 and 1, respectively) were omitted from statistical testing. In case of rejection of the null hypothesis ($\alpha \leq 0.05$), the Wilcoxon rank-sum test was employed for pairwise Bonferroni-adjusted post hoc tests to identify all federal states differing significantly from Rhineland Palatinate. We assessed the correlation between pairs of variables with non-directional Spearman's rho. Note that some relationships (e.g. decreasing DF for the regional scenarios and for increasing the number of upstream WWTPs) were mathematically evident and required no statistical testing.

3. Results

3.1. Dilution factors for the local scenario ($DF_{i,0}$)

For the national dataset, 60% of the $DF_{i,0}$ fell below the fixed factor of 10 (median $DF_{i,0} = 5$). Similarly, a substantial proportion of $DF_{i,0}$ for the federal state dataset (~40%, median $DF_{i,0} = 14.5$) fell below 10 based on

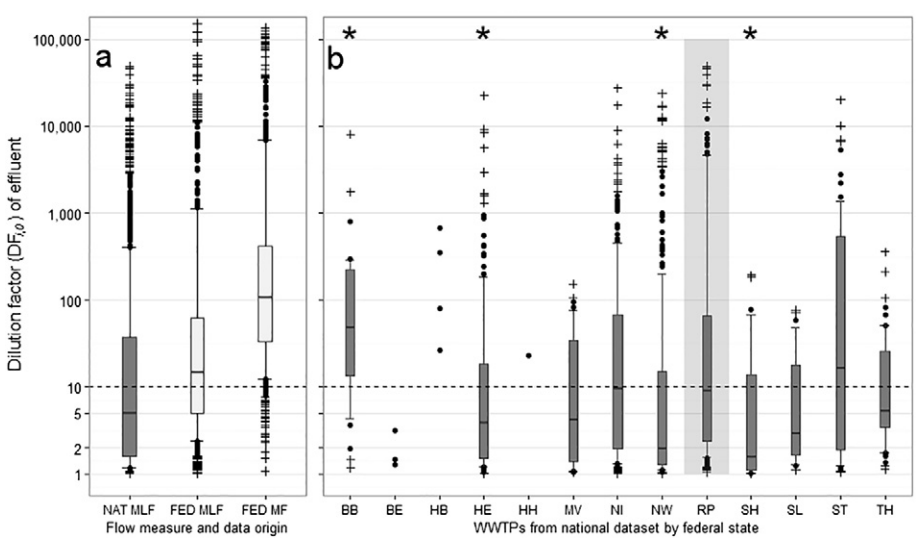

Fig. 2. a) Distribution of calculated $DF_{i,0}$ for the national dataset (NAT MLF, 1225 records) and federal dataset (FED MLF and FED MF, 675 records each). 11 values with $DF_{i,0} > 150{,}000$ are not shown (FED MLF: 3 values; FED MF: 8 values). b) $DF_{i,0}$ of national dataset grouped by federal state. Upper and lower error bars indicate 90th and 10th percentile, respectively. Boxes include 25th to 75th percentile with median as horizontal black line. Dashed line indicates DF_{fix} of 10. Crosses indicate values < 5th and >95th percentile. Asterisks indicate federal states with statistically significant different $DF_{i,0}$ (all $p < 0.05$) compared to RP (shaded area).

MLF, whereas only 8% of $DF_{i,0}$ (median $DF_{i,0} = 106$) fell below a factor of 10 based on MF (Fig. 2a). When grouping the national data by federal state, the median $DF_{i,0}$ from eight out of 13 states fell below 10 (Fig. 2b). Please note that all three WWTP of Berlin (with PE > 500,000) even fell below a DF of 5. The $DF_{i,0}$ for RP from the federal dataset (Fig. 2a), which included WWTPs of all sizes, were overall higher than for WWTPs from RP from the national dataset (Fig. 2b, RP), which was limited to large WWTPs. The difference between $DF_{i,0}$ for the datasets was particularly prominent for the lower (25%) percentile (Federal state dataset: <5, national dataset: <2.5). For at least 10% of water bodies the $DF_{i,0}$ was ≤2.4 (both datasets).

The $DF_{i,0}$ differed significantly among states ($p < 0.001$, KW-test). Pairwise post-hoc tests (Bonferroni-adjusted Wilcoxon rank-sum tests) showed that the $DF_{i,0}$ of Brandenburg ($p = 0.048$, $n = 36$), Hesse ($p = 0.003$, $n = 173$), North Rhine-Westphalia ($p < 0.001$, $n = 380$) and Schleswig-Holstein ($p = 0.002$, $n = 41$) differed significantly from the $DF_{i,0}$ of Rhineland Palatinate ($n = 157$).

$DF_{i,0}$ below 10 occurred primarily in streams with flow rate below 1 $m^3 s^{-1}$, irrespective of the flow rate measure (i.e. MF or MLF; Fig. 3).

Under MLF conditions, $DF_{i,0}$ below 10 can be observed even for very small WWTPs (<1000 PE) (Fig. 3b), whereas under MF conditions mainly larger WWTPs result in $DF_{i,0}$ below 10 (Fig. 3c). For the national dataset, WWTP size in terms of PE correlated only weakly with flow rate (non-directional Spearman's rho for MLF = 0.27), whereas correlation between these variables was higher for the federal dataset (non-directional Spearman's rho for MLF = 0.61 and for MF = 0.52). Lowering the DF_{fix} from 10 to 5 would increase the amount of sites with empirical $DF_{i,0} > DF_{fix}$ to 50% and 75% for the national and the federal dataset, respectively (Fig. 4a and b). For the federal dataset, a DF_{fix} of 2 covers at least 90% of the calculated $DF_{i,0}$, whereas for the national dataset the same observation applies for a DF_{fix} of 1.2.

3.2. Dilution factors for the regional scenario ($DF_{i,n}$)

For the regional scenario (II), the $DF_{i,n}$ of 70% and 60% of sites, for the national and federal dataset, respectively, fell below the DF_{fix} of 10, based on MLF. Under MF conditions in the federal dataset still 10% of the $DF_{i,n}$ fell below 10. Overall, a 1.5-fold drop of median $DF_{i,n}$ could be observed compared to the local median $DF_{i,0}$ (national dataset, median $DF_{i,n}$: 3; federal dataset under MLF, median $DF_{i,n}$: 7.3; federal dataset under MF, median $DF_{i,n}$: 55.5 Fig. S1). Lowering the DF_{fix} from 10 to 5 would increase the amount of sites with a $DF_{i,n} > DF_{fix}$ to 40% and 60% for the national and the federal dataset, respectively (Fig. 4a and b).

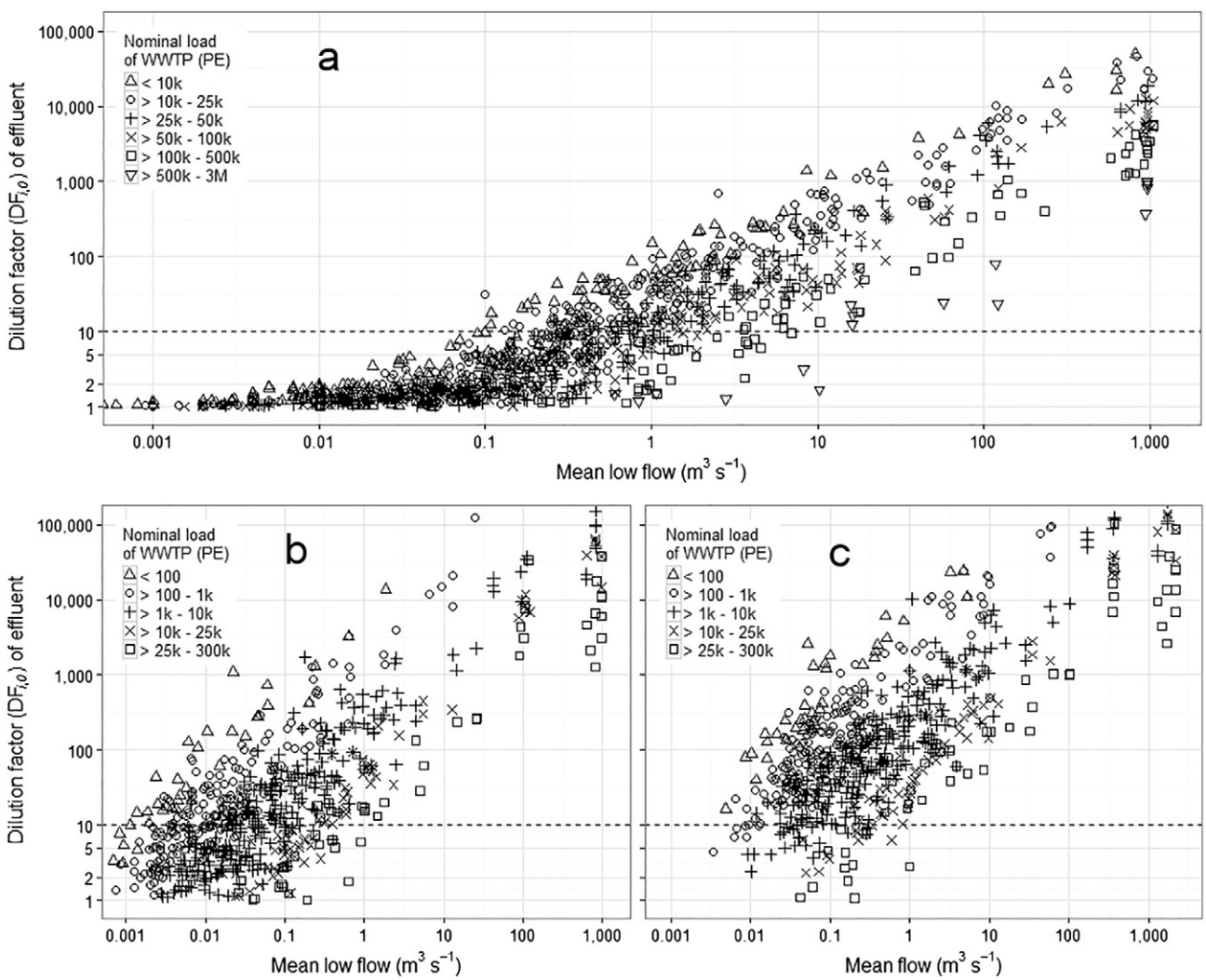

Fig. 3. Local dilution factor ($DF_{i,0}$) against flow rate of the respective streams, symbols indicate size of WWTP (nominal load in PE). a) For national dataset with MLF, 11 values where MLF < 0.0005 not shown. b) For federal dataset with MLF, 5 values not shown (2 values where MLF < 0.0005 and 3 values where $DF_{i,0} > 150{,}000$). c) For federal dataset with MF, 8 values not shown, where $DF_{i,0} > 150{,}000$. Dashed line indicates DF_{fix} of 10.

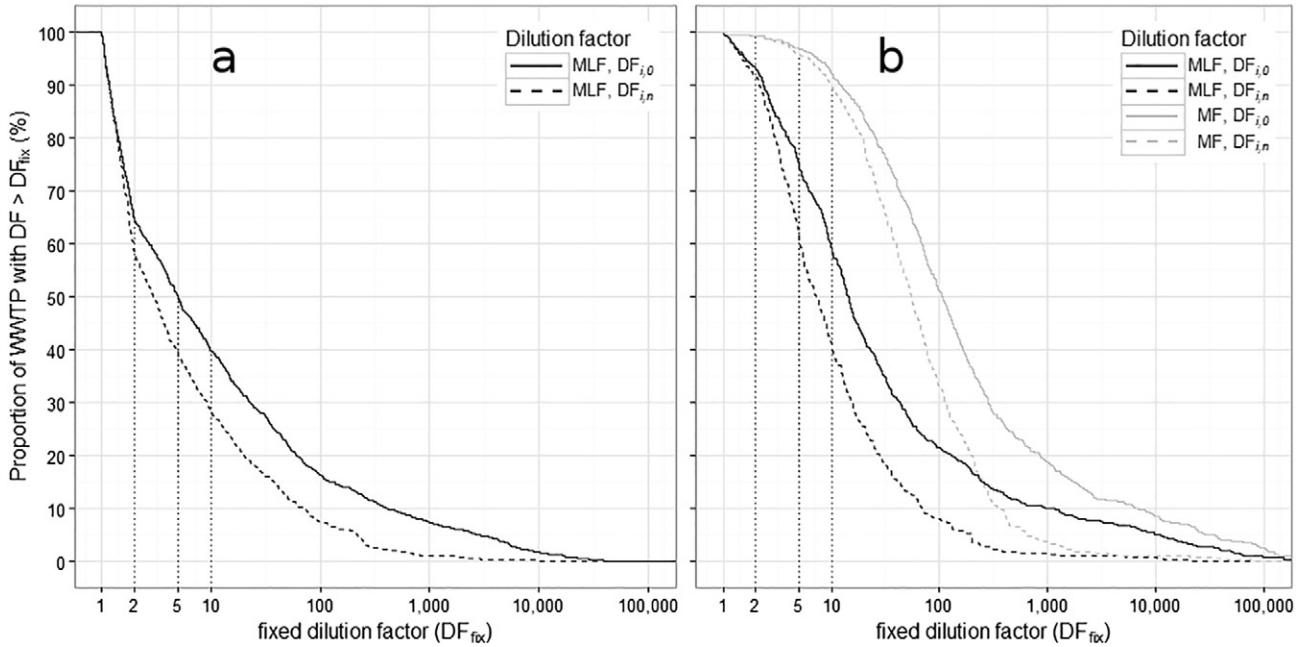

Fig. 4. Percentage of sites with DF > DF_{fix}, corresponding to different DF_{fix} for the national dataset (a) and the federal dataset (b). a) The continuous line indicates the local approach ($DF_{i,0}$), the dashed line indicates the regional approach ($DF_{i,n}$), both for MLF. b) The continuous lines indicate the local approach ($DF_{i,0}$), the dashed lines indicate the regional approach ($DF_{i,n}$), in black under MLF, in grey under MF.

3.3. Dilution factors for the stepwise regional scenario ($DF_{down,j}$)

$DF_{down,j}$ for two groups of stream sizes (0.1 $m^3\ s^{-1}$ < MLF ≤ 1 $m^3\ s^{-1}$ and 1 $m^3\ s^{-1}$ < MLF ≤ 10 $m^3\ s^{-1}$) decreased with the number of WWTPs included along the stream (or catchment for the federal dataset, Fig. 5a and b). The $DF_{down,j}$ exhibited the largest drop (approx. 2.5 fold for median) from first to second WWTP but quickly levelled off afterwards. Smaller streams of both datasets were associated with smaller $DF_{down,j}$, as for the local scenario.

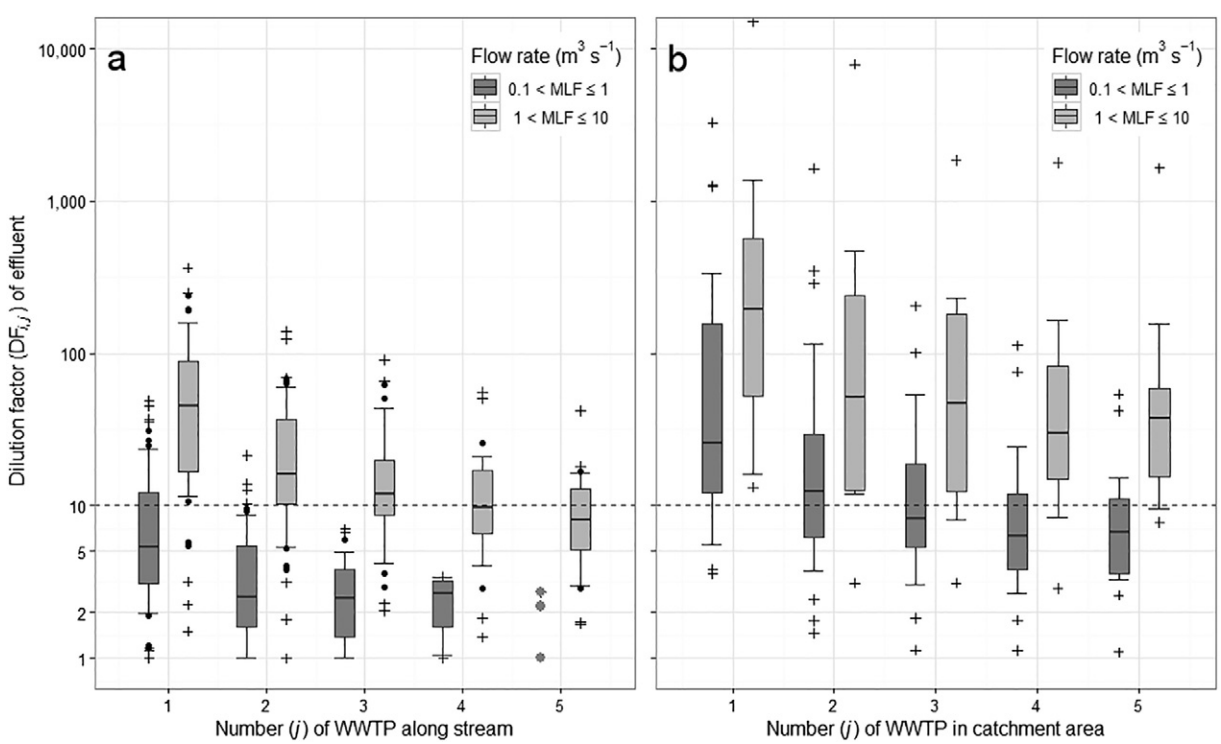

Fig. 5. Association between $DF_{down,j}$ and the number of upstream WWTPs as well as the respective flow rate class, for the German dataset (a) and the Rhineland Palatinate dataset (b). "+" indicate values < 5th and >95th percentile, respectively. Upper and lower error bars indicate the 90th and 10th percentile, respectively. Boxes include 25th to 75th percentile with median as horizontal black line. Dashed horizontal line indicates DF_{fix} of 10.

4. Discussion

4.1. Local dilution factors

The range of $DF_{i,0}$ calculated for the federal dataset based on MF (10th and 90th percentile of $DF_{i,0}$: 12 and 6926) is in general agreement with median DF for creeks, streams and large rivers (8, 34 and 3,200, respectively) modelled for the Netherlands (De Greef and De Nijs, 1990). Furthermore, this study also supports our finding that $DF_{i,0}$ below 10 occur especially in smaller receiving waters. Keller et al. (2014) derived an annual median DF of 32 for Germany based on modelled effluent volumes and the annual variability of flow rate. This value is six fold and two fold larger than our median $DF_{i,0}$ for the national and the federal dataset based on MLF (i.e. 5 and 14.5, respectively). This discrepancy may be explained by the fact that the derived annual median DF is based on the seasonal variability of flow rate, whereas our study calculated $DF_{i,0}$ for MLF conditions. Consequently, $DF_{i,0}$ below the derived annual median DF are reasonable.

Given the calculation of the $DF_{i,0}$ (see Eqs. (1) and (2)), flow rate (Q) is a main driver of $DF_{i,0}$. However, the influence of the stream flow rate on the $DF_{i,0}$ could be balanced if *EV* would be correlated with stream size, for example if larger WWTPs would mainly discharge into larger streams. However, for the national dataset, we found only a weak correlation between WWTP size in terms of nominal hydraulic load and flow rate of receiving streams (Spearman's rho = 0.27). In fact, 70% of WWTPs with nominal loads up to 100,000 PE discharged into rather small streams with MLFs below 1 $m^3\ s^{-1}$ (Fig. 3a). In extreme cases such as for the federal state of Berlin, streams with MLF well below 10 $m^3\ s^{-1}$ receive effluents from WWTPs larger than 500,000 PE. Overall, a high fraction of 87% of $DF_{i,0}$ fell below the DF_{fix} of 10 for receiving water bodies with MLF < 1 $m^3\ s^{-1}$. The low correlation can partly be explained by the lower gradient in WWTP sizes, given that the correlation was higher for the federal dataset (Spearman's rho = 0.61). However, the correlation was similarly low if only WWTPs > 10,000 PE were considered for the federal dataset (Spearman's rho = 0.23). Notwithstanding, even for the full federal dataset, lower flow rate of small streams is not strongly associated with lower effluent volumes, which means that in general small streams are at highest risk of low dilution (Fig. 3). Similarly, studies in Switzerland and Spain found DF < 10 for streams with flow rates < 10 $m^3\ s^{-1}$, and generally, across Europe, DF decreased with decreasing stream size, though the DF reported in these studies refer to short periods of stream flow conditions and originate from a low number of sampling sites (Baker and Kasprzyk-Hordern, 2013; Gros et al., 2010; Kasprzyk-Hordern et al., 2009; Ort and Siegrist, 2009; Verlicchi et al., 2014; Vermeirssen et al., 2006). Thus, our large scale study confirms these observations.

The vulnerability of small streams may be exacerbated by the fact that extreme weather and hydrological conditions (i.e. drought) that are spatially limited to a certain region particularly affect small streams in the corresponding area, whereas large streams may be less affected because of larger catchments extending beyond the affected region. Given an increase in extreme weather events with climate change (Stocker et al., 2013), with predicted reductions in summer rainfall of approximately 40% until 2100 for some regions of Austria and Germany (e.g. Alps, Upper Rhine Valley) (Leipprand et al., 2008; Schöner et al., 2011), the frequency of streams and periods with low flow, and hence low dilution potential, may increase.

For both datasets a large fraction of streams (40% and more) exhibited a $DF_{i,0}$ below 10 under mean low flow conditions. To which extent this result can be extrapolated to other European regions remains uncertain. The national dataset was censored to WWTPs with nominal loads ≥ 10,000 PE and the fraction of streams with low $DF_{i,0}$ is consequently higher than for the total population of streams with a WWTP. In addition, our national scale analysis did not cover two of the largest Southern German states. Based on the national dataset, the results for Rhineland Palatinate differed from those of other federal states. Notwithstanding, the median $DF_{i,0}$ were similar to those of Rhineland Palatinate and for states that exhibited significant differences, only Brandenburg had a higher median $DF_{i,0}$. Thus, in general our results for the federal state data are likely to underestimate the fraction of streams with $DF_{i,0} < DF_{fix}$ in other federal states.

4.2. Regional dilution factors

The exposure assessment according to the EMA guideline (EMA, 2006) focuses on a standard local WWTP emission scenario, whereas the emissions from upstream WWTPs are ignored. Nevertheless, they might contribute to local concentrations in case of (1) no decay of a substance or (2) pseudo-persistence, i.e. when multiple WWTPs along a receiving stream or in an upstream catchment lead to input rates exceeding degradation rates. Indeed, accumulation of consecutive effluent inputs by several WWTPs along a stream have been documented (Baker and Kasprzyk-Hordern, 2013). About 60% of the WWTPs in the national and federal dataset had at least one additional WWTP located further upstream. Therefore, a regional approach to calculate $DF_{i,n}$ represents a realistic (worst case) scenario. We found median $DF_{i,n}$ of 3 and 7.3 for the national and the federal dataset, respectively. To our knowledge only one other study (Kapo et al., 2015) modelled DF with a regional scenario, but in contrast to our study DF were calculated based on single compound concentrations in the effluents and the receiving water. Even though the concentration based approach on DF does not need *EV*, the results are expected to be of similar accuracy compared to our results. This study reported a similar median regional DF of 5 for substances with no decay rate during low flow conditions for streams in the US. For MF, we calculated a median regional $DF_{i,n}$ of 55 for the federal dataset that is approximately half of that reported for the US dataset (median DF = 105 for no decay). This difference is in a similar order of magnitude as for the median $DF_{i,0}$ between the German federal states (Fig. 2b). Still our calculated $DF_{i,n}$ are likely overestimated, particularly for large streams, because these often had upstream catchments extending federal and national borders and we lacked data on *EV* for these areas. Hence, the underestimation of total *EV*, likely resulted in an overestimation of the $DF_{down,j}$, for the streams crossing administrative borders.

We also calculated stepwise regional $DF_{down,j}$ to illustrate the influence of the number of upstream WWTPs on the DF. Here, we only used streams with MLF from 0.1 $m^3\ s^{-1}$ up to 10 $m^3\ s^{-1}$ given the data issues for large streams crossing administrative borders outlined above. For these streams we observed a 2.5 fold drop in the median $DF_{down,j}$ as well as in the 5th and the 25th percentiles. Consideration of additional *EV*s from WWTPs further upstream led only to negligible reductions in median $DF_{down,j}$.

Thus, our results suggest that considering regional exposure routes may be important if the aim is to capture worst case conditions, while it may be sufficient to include only one upstream WWTP along a stream or in a catchment. This conclusion is in agreement with the study on US streams demonstrating that median regional DF responded negligibly to the differences in degradation half-lifes ranging from 1 h to 24 h (Kapo et al., 2015), where higher degradation half-lifes would correspond to longer distances between successive effluent input points along a stream.

4.3. General uncertainties of our analysis

Several uncertainties should be considered when interpreting our results. First, we converted the *EV* from $m^3\ year^{-1}$ to $m^3\ s^{-1}$ to match the measure of stream flow rate. However, this approach disregards daily and seasonal differences in effluent volumes as observed in previous studies, where low flow periods might also overlap with lower *EV* (Kasprzyk-Hordern et al., 2009; Ort and Siegrist, 2009). As a result, DF based on MLF in our study might underestimate the real DF during low flow periods, if lower *EV* overlaps with low flow periods. In addition

to that, temporal and spatial changes have also been observed for contaminant loads, with lower loads of antibiotics and β-blockers during summer time (Munz et al., 2016; Roberts et al., 2016; Vieno et al., 2007). Moreover, the stream flow rate varies inter-annually (Wagner and Schmidt, 2004), including variations in the duration and frequency of water levels below the MLF. Data with higher temporal resolution on *EV* would allow for the calculation of more reliable DF, when matching with stream flow rates for low flow periods within a year. However, such data were not available for large scale analyses. In fact, regulatory bodies therefore have introduced the measure of MLF to account for the assessment of frequently occurring low flow conditions. The federal dataset covered different decades, where information on MF, MLF and *EV* were collected between 1979 and 1998 but information on WWTP sizes and PE were only available for 2010. Whereas the population size can be considered relatively stable for the study region, changes in water consumption may have occurred as well as changes in MF and MLF given climate change. Here decreasing water consumption would lead to an underestimation of DF, whereas decreasing MLF would lead to a overestimation. Changes in the water consumption per capita are very likely the dominating variable concerning these uncertainties. However, related uncertainties can be considered minor compared to other uncertainties outlined above.

4.4. Evaluation of regulatory assumptions

The REACH guidance document (ECHA, 2016) bases the environmental exposure assessment of chemical emissions on a standard WWTP with a nominal load of 10,000 PE and an *EV* of 0.023 $m^3\ s^{-1}$ (2000 $m^3\ d^{-1}$). In Germany, these larger WWTP with PE > 10,000 cover >90% of the total hydraulic wastewater load, albeit they represented only 19% of the number of all WWTPs in the federal state dataset (total of 678 WWTPs). The national dataset consisted of 1042 WWTP with PE > 10,000. However, a considerable fraction (25%) of them was at least five-fold larger than 10,000 PE, corresponding to a much higher pressure on the receiving water bodies (Fig. S2). Corresponding receiving waters with flow rates of at least 0.23 $m^3\ s^{-1}$ (including the *EV*) are assumed under the DF_{fix} of 10. Our analysis revealed, however, that 50% and 72% of WWTPs in the national and the federal dataset were adjacent to receiving waters with much smaller MLF, respectively (Fig. S3). Thus, the assumed stream flow rate used in the exposure assessment is generally overestimated and the environmental concentrations of substances approved by related risk assessments can be much higher than expected under the DF_{fix}. Additionally, on-site exposure assessments assume a MF/MLF ratio of 3, if MLF rates are not available. By contrast, the ratio of MF/MLF was larger than 3 for 79% of receiving waters of the federal state dataset (median: 6; lower quartile: 3.85; upper quartile: 10.29, see Fig. S4). Consequently, the actual MLF of these receiving streams is smaller and the assumed ratio used in on-site exposure assessments overestimates the real flow conditions, what might in turn result in contaminant concentrations above predicted environmental concentrations. These findings are of particular importance for Southern Europe, where the ratio of MF to MLF is even larger than in Central Europe (Dettinger and Diaz, 2000; Keller et al., 2014).

4.5. Implications for chemical risk assessment

Despite uncertainties, the DF_{fix} of 10, as e.g. used in the EMA-guideline (EMA, 2006) or the ECHA-guidance document (ECHA, 2016), overestimates the dilution potential for a large amount of stream sites from both datasets under the regulatory accepted MLF conditions. Calculated $DF_{i,0}$ for 50% and 25% of national and federal sites under MLF conditions were half of the DF_{fix} (i.e. 5 or lower), respectively. Applying the DF_{fix} of 10 in standard exposure assessments therefore leads to an underestimation of environmental concentrations of pharmaceuticals, PCP, biocides or industrial chemicals under the assumed worst-case conditions. Indeed, measured environmental concentrations (MEC) exceeding PECs were found in a recent study (Meyer et al., 2016). They calculated minimum and maximum PECs for carbamazepine, diclofenac and metoprolol in German rivers using two models suggested in the EMA recommendations (EMA, 2006), of which both implemented the DF_{fix} of 10. Almost 50% of MECs for diclofenac exceeded the maximum PEC and the MECs exceeded the majority of minimum PECs for carbamazepine and metoprolol. Besides the ratio of patients using these pharmaceuticals, the DF_{fix} of 10 could have contributed to the potential underestimation of chemical risks in most sites in their study and this underestimation may also apply to other European countries. If the exposure assessment shall continue to be based on a DF_{fix} and a potential underestimation of risks for the majority of streams is considered unacceptable, adjustments to a lower DF_{fix} in the range of 2 would be required. Moreover, detailed information on effluent composition would be crucial, since the type of sewage system (combined or separated sewer systems for rainwater) affects contaminant concentrations in the effluents and the DF of effluents (Ort and Siegrist, 2009). If information on the effluent composition is available, this would also allow consideration of potential mixture effects (Neale et al., 2017). Besides that, regional exposures from multiple sources due to environmental persistence or pseudo-persistence of contaminants would require consideration, too. Finally, annual stream flow rates have decreased regionally during the last century, resulting in generally lower MLF, especially in Southern and Eastern Europe and climate change may require re-evaluations of the DF_{fix} over time (Stahl et al., 2010). But even if all sites met the standard DF_{fix} of 10, the concept of a fixed dilution factor during exposure assessment would still be a matter of concern, because several WWTP effluents in the United Kingdom would need DF between 10 and 1000 to meet 2012 proposals for environmental quality standards (EQS) for several emerging pollutants (EC, 2012; Gardner et al., 2012).

5. Conclusion

Our analysis represents the first German-wide assessment of empirical DF for effluents of municipal WWTPs. $DF_{i,0}$ for a substantial amount of sites in both datasets fell below 10, showing that the DF_{fix} of 10 employed in the exposure assessments of many chemical regulations does not capture the smaller dilution potential of most receiving waters during low flow conditions. 50% of $DF_{i,0}$ from the national dataset were smaller than 5. This results in a two-fold lower predicted environmental concentration, which may contribute to underestimating environmental concentrations. The nearest upstream WWTP has the biggest impact on $DF_{down,i}$, whereas more distant effluent inputs are negligible for local risks. Reducing the standard regional DF_{fix} to 2 would double the percentage of sites in both datasets exceeding this dilution factor, corresponding to at least 60% and 90% of respective stream sites in the national dataset and the federal dataset under MLF. If the aim is to decrease risks for surface water quality and for failing proposed EQS criteria, freshwater managers should consider revising the DF_{fix}, potentially also considering (1) effluent composition and type of respective sewage systems (2) regional exposure from multiple effluent inputs, and (3) decreasing annual flow rates of receiving waters in some regions.

Disclaimer

The views expressed in this article are the personal views of the authors and may not be understood or quoted as being made on behalf of or reflecting the position of the German Federal Environment Agency.

Conflict of interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

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